

Let ViT Speak: Generative Language-Image Pre-training

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Abstract

In this paper, we present **Generative Language-Image Pre-training** (GenLIP), a simplified generative pretraining approach for Vision Transformers (ViTs) tailored to multimodal large language models (MLLMs). To better align ViTs with the autoregressive nature of LLMs, GenLIP trains a ViT to predict language tokens directly from visual tokens using a standard language modeling objective—without contrastive batch construction or an additional text decoder. GenLIP offers three key advantages: (1) **Simplicity**: a single transformer jointly models visual and linguistic tokens; (2) **Scalability**: it scales efficiently with both data and model size; (3) **Performance**: it achieves competitive or superior results across diverse multimodal benchmarks. With training totally 8B samples on Recap-DataComp-1B, GenLIP matches or surpasses strong baselines such as SigLIP2. With continued pretraining on multi-resolution images at native aspect ratios, GenLIP further excels at detail-sensitive tasks such as OCR, chart understanding, and visual question answering, making it a strong foundation for vision encoders in MLLMs.

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Project Page: <https://pixelfang.github.io/vitspeak.github.io>

1 Introduction

Multimodal Large Language Models (MLLMs) have emerged as a transformative paradigm in artificial intelligence, demonstrating remarkable capabilities in understanding and reasoning across vision and language modalities [6, 12, 43, 60, 83]. The prevailing architecture of MLLMs comprises three core components: a vision encoder for processing visual information [13, 21, 56, 78], a connector for bridging modalities, and a large language model (LLM) as the reasoning engine [1, 5, 64, 66]. Among these components, the vision encoder serves as the *perceptual foundation*, responsible for extracting meaningful visual representations that can be effectively consumed by the downstream LLM. The quality and design of this vision encoder fundamentally determine the upper bound of an MLLM’s visual understanding capabilities. Currently, large-scale Vision-Language Pre-training (VLP) on billions of image-text corpora have become the dominant approach for developing strong vision encoders.

Contrastive learning based VLP methods, exemplified by CLIP [56] and SigLIP [78], are among the most widely adopted vision encoders for MLLMs [9, 62, 63]. These methods typically employ a dual-encoder architecture to encode each modality separately and align them using a contrastive objective. However, contrastive pretraining introduces an *objective mismatch* with the generative nature of MLLMs: contrastive

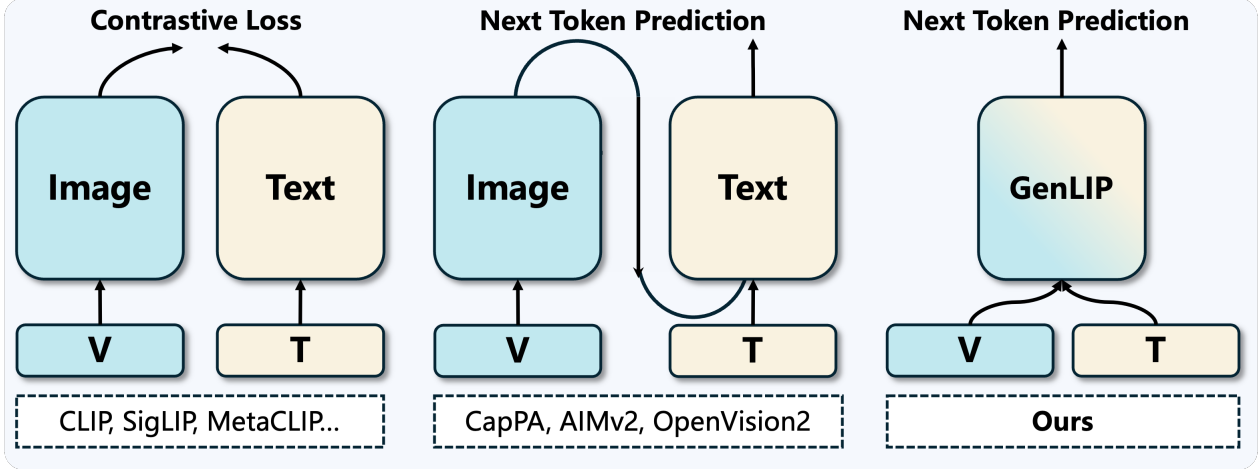


Figure 1 Unlike previous vision-language pretraining methods of complex two-tower design, our proposed GenLIP is with great simplicity in architecture design. In this figure, we use “V” and “T” to denote visual and textual inputs.

learning encourages discriminative representations, whereas MLLMs are optimized for autoregressive token prediction.

Another stream of works focus on generative pretraining, such as CapPa [67], AIMv2 [23], and OpenVision2 [46]. These methods typically cascade the vision encoder with a text decoder, and train the cascaded model using an autoregressive language modeling objective. In this setup, the vision encoder is updated indirectly via gradients flowing through the text decoder. Related hybrid designs, such as CoCa [77] and SigLIP2 [68], further introduce an additional text encoder to combine contrastive and generative objectives. While these approaches narrow the gap, the redundant architecture design and indirect optimization can limit efficiency for training a scalable vision encoder in MLLMs.

To unleash the full potential of generative vision-language pretraining, we advocate for a minimalist design philosophy that eschews unnecessary complexity while enabling efficient scalability. In this work, we propose a simplified framework for generative vision-language pretraining: **Generative Language-Image Pretraining** (GenLIP), a minimalist yet scalable framework that departs from the complex designs of prior VLP methods. Instead of introducing novel architectural components, our core insight is elegantly simple: *let the Vision Transformer (ViT) speak directly*—requiring no contrastive batch construction and no additional text module.

Instead of indirectly optimizing vision transformers through auxiliary text modules, GenLIP directly pretrains the ViT to generate language tokens that describe the visual content using only a standard autoregressive language modeling objective.

This pure minimalist generative paradigm naturally aligns the vision encoder with the way LLMs operate, mitigating the objective mismatch in contrastive approaches while simplifying designs and improving efficiency for generative frameworks. GenLIP’s design philosophy offers several compelling advantages: **(1) Simplicity:** GenLIP uses a minimalist form of a single vision backbone and an autoregressive objective, without contrastive losses or additional text modules; **(2) Scalability:** GenLIP scales effectively with data and model size, yielding strong empirical performance. **(3) Performance:** GenLIP achieves competitive performance as vision encoder for MLLMs, especially for optical character recognition (OCR) tasks. Together, these results highlight GenLIP, a minimalist generative pretraining, as an effective and scalable path toward building strong vision encoders for MLLMs.

In summary, this work introduces GenLIP, a minimalist framework toward efficient and scalable generative vision-language pretraining. Our extensive experiments demonstrate that this principled, end-to-end approach not only exhibits remarkable scalability but also achieves competitive or superior performance under the evaluated MLLM settings. We believe these findings chart a more direct, efficient, and scalable course for developing powerful vision-language models.

2 Related Work

The convergence of computer vision and natural language processing has been driven by large-scale vision-language pretraining, which aims to learn robust, generalizable multimodal representations from massive image-text corpora. Typical VLP methods can be grouped into three categories based on architectural design and training objectives: dual-encoder contrastive pretraining, encoder-decoder generative pretraining, and simplified single-transformer pretraining.

Dual-Encoder Contrastive Pretraining. A broad line of research has investigated Contrastive Language-Image Pretraining. CLIP-style architectures [13, 14, 31, 56, 74, 78] are fundamentally based on a dual-encoder (two-tower) design, which learns to align image and text representations within a shared embedding space using an InfoNCE or similar contrastive objective. Subsequent works improve alignment by leveraging high-quality image-text pairs [15, 22, 25, 33, 41, 76, 81] or dense region-level captions [39, 42, 79] for fine-grained representation learning. While effective for discriminative tasks such as classification and retrieval, contrastive pretraining primarily focuses on global alignment and does not facilitate deep cross-modal interaction.

Encoder-Decoder Generative Pretraining. To enable richer cross-modal reasoning, recent works [3, 23, 46, 69, 71] adopt generative pretraining, typically cascading a vision encoder with a text decoder. For example, Aimv2 [23] couples a vision encoder with a multimodal decoder that autoregressively generates raw image patches and text tokens, whereas CapPa [67], GIT [69] and OpenVision 2 [46] stack a text decoder on top of the image encoder and pre-train the model using only a captioning loss. Most recently, some studies [37, 38, 41, 45, 68, 77] form hybrid pretraining schemes that combine a contrastive dual-encoder for image-text alignment with a generative decoder for captioning.

Discussion. Despite their success, existing methods often rely on multiple towers or multiple optimization objectives, which increases model complexity and limits efficiency. Moreover, alignment is often performed at later stages rather than within the image encoder itself, which can constrain early cross-modal interactions. Different from these works, we propose a minimalist generative vision-language pretraining framework with simplified architecture and training objective—a single transformer and a single language modeling objective.

Single-Transformer Pretraining. Recently, some works also explored vision-language pretraining under a simplified single-Transformer architecture with different objectives. Among them, SuperClass [28] proposes vision transformer pretraining with a single Transformer tower using token-level classification targets. VL-BEiT [8] and OneR [30] aim to unify vision-language representation learning within a single-tower Transformer, but still rely on multiple objectives. Beyond vision transformer pretraining, several recent efforts [11, 18–20, 34, 61] aim to build native MLLMs with a single transformer and a single language modeling objective.

Discussion. In particular, GenLIP is architecturally close to SAIL [34], as both use a single transformer with a language modeling objective. However, SAIL focuses on building a native MLLM with a simplified architecture based on pretrained LLMs, whereas GenLIP is designed to pretrain a scalable vision encoder from scratch to better serve modular MLLMs [7, 12, 35]. This distinct goal also leads to different design choices.

3 Approach

This section details GenLIP, our minimalist implementation for generative vision-language pretraining. We first introduce the core designs of our approach, including the model architecture, data representation, and training objective, all designed for minimalist generative vision-language pretraining. We then provide pretraining details, including pretraining datasets and training schedule.

3.1 GenLIP Framework

Instead of introducing novel architectural components, GenLIP is built upon a minimalist unified modeling paradigm for vision encoder pretraining. Specifically, we build GenLIP with a simple transformer architecture in the spirit of *let the ViT Transformer speak directly*, analogous to how LLMs generate text. We keep all designs simple, introducing only minimal but necessary modifications for improving representation.

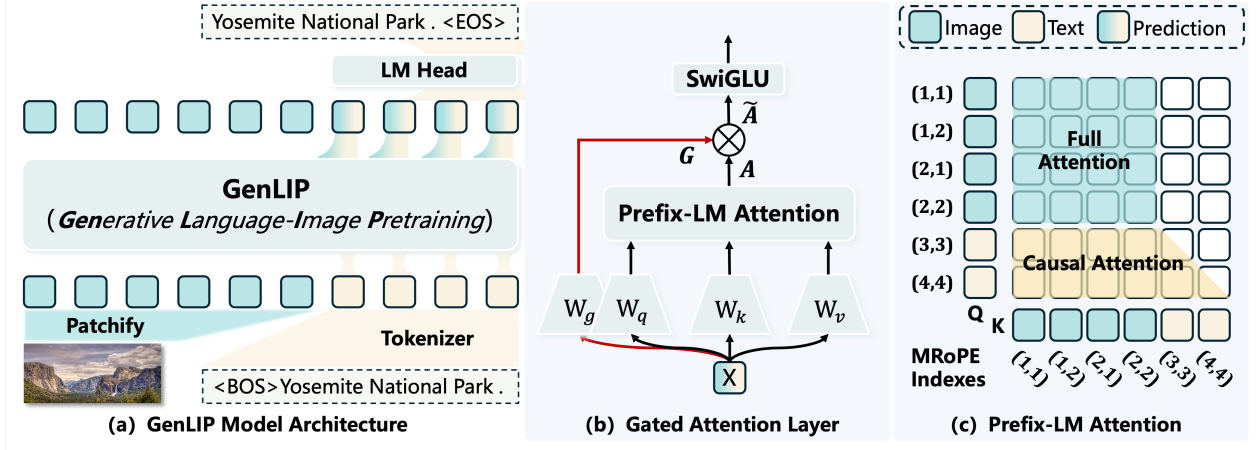


Figure 2 An overview of the GenLIP framework for minimalist generative vision-language pretraining. **(a) GenLIP Model Architecture:** a single Transformer architecture to process a concatenated visual-prefix sequence. The next token prediction is performed exclusively on text tokens via a language modeling head. **(b) Gated Attention Layer:** our basic layer unit of GenLIP, read line in the figure is the forward path of gating signals, which is then element-wise multiplied with the attention output to control the information flow. **(c) Prefix-LM Attention Mechanism:** allows both bi-directional attention for image tokens and causal attention for text tokens. Multimodal Rotary Position Encoding (MRoPE) injects position information into the query (Q) and key (K) vectors.

Data Format. All pretraining data for GenLIP is structured as image-text samples, denoted as $\{(I_i, T_i)\}_{i=1}^N$, where each image I_i is associated with its caption T_i . Each image I_i is partitioned into a sequence of non-overlapping patches $\{v_0, v_1, \dots, v_M\}$ using a convolutional patch embedding layer, as in standard ViT models. The corresponding text T_i is tokenized into a sequence of subword tokens $\{t_0, t_1, \dots, t_L\}$ using an off-the-shelf text tokenizer (Qwen3 [75]). The resulting image patch embeddings and text token embeddings are concatenated into a single sequence, with the image embeddings preceding the text embeddings. The final input sequence S for a given pair (I_i, T_i) is:

$$S = [v_0, \dots, v_M, t_0, \dots, t_L]. \quad (1)$$

Architecture. The architecture of GenLIP follows simplicity and effectiveness, centered around a unified Transformer encoder that processes a concatenated sequence of image and text tokens. As illustrated in Figure 2, the model consists of three components: modality-specific embedding layers, a unified Transformer with a prefix-LM attention implementation, and a language modeling (LM) head for token prediction.

To enable effective cross-modal interactions and unified modeling of the concatenated visual-prefix multimodal sequence, we make two small but crucial modifications to a standard Transformer. (i) To better encode the position information in a concatenated visual-prefix multimodal sequence, we use multimodal rotary position encoding (MRoPE) [70] and discard the absolute position embeddings for image patches. (ii) We replace the basic full attention with prefix-LM attention [57] in all transformer blocks, where image tokens attend bidirectionally and text tokens attend causally. Based on the above two modifications, we directly apply the GenLIP architecture to process the unified multimodal sequence, without additional modality-specific designs in the network architecture.

Objective. GenLIP adopts a single standard autoregressive language modeling objective, applied exclusively to the textual part of the sequence. The model is trained to predict the next text token conditioned on the preceding image tokens and text tokens, directly models the conditional probability distribution $P(T|I)$. The objective is to minimize the negative log-likelihood of the text sequence:

$$\mathcal{L}_{\text{LM}} = - \sum_{k=0}^L \log P(t_k | \{v_j\}_{j=0}^M, \{t_i\}_{i=0}^{k-1}; \theta) \quad (2)$$

where θ denotes the model parameters to be optimized, and $P(t_k | \{v_j\}_{j=0}^M, \{t_i\}_{i=0}^{k-1})$ is the predicting probability of k -th text token conditioned on all preceding visual and textual tokens.

Using GenLIP as a Vision Encoder. When using GenLIP as a vision encoder, we extract the visual embeddings from the output of the last transformer block and feed them into a lightweight MLP projector to align with the input space of the LLM. In this process, the language modules of GenLIP (tokenizer and LM head) are discarded, all other components are retained and directly used. The Prefix-LM attention mechanism is degraded into a standard full attention for visual modeling when used as vision encoder.

3.2 Gated Attention

While the above unified architecture is effective for generative vision-language pretraining, we observe a notable side effect: attention becomes overly concentrated on the first token of the input sequence, a phenomenon known as the *attention sink*. This issue is particularly pronounced in our mixed-modality setting, as shown in Figure 3. Under full attention, certain visual tokens can freely aggregate global information from all patches, effectively becoming image-level summaries. Since text tokens only access visual information through causal attention over this shared visual prefix, the model learns a shortcut: compressing visual information into a few sink tokens for efficient language prediction, at the cost of degrading spatial diversity in visual representations. Consistent with findings in [54], this leads to (i) obvious loss spikes during pretraining, and (ii) attention distributions where the first token absorbs most of the attention mass, reducing the effective utilization of visual tokens. As a result, the pretrained ViT fails at discriminative tasks such as ImageNet linear probing and exhibits unstable scaling behavior—both undesirable for our target usage as a vision encoder for MLLMs.

Inspired by [54], we introduce a gated attention mechanism to regulate information flow in the mixed-modality modeling space. Given input hidden states $X \in \mathbb{R}^{n \times d}$ for a Transformer block, we compute a standard attention output $A = \text{Attn}(X)$ and apply an input-dependent gate:

$$G = \sigma(XW_g + b_g), \quad \tilde{A} = G \odot A, \quad (3)$$

where $\sigma(\cdot)$ is the sigmoid function, W_g and b_g are learnable parameters, and $\odot$ denotes element-wise multiplication. The gated attention output $\tilde{A}$ is then used in the standard residual pathway. By modulating attention outputs on a per-token basis, the gate prevents text tokens from collapsing their attention onto a small subset of visual tokens and encourages the model to leverage spatially distributed visual features. In practice, gated attention alleviates loss spikes, accelerates convergence, and stabilizes scaling behavior (Figure 6).

3.3 Pretraining Details

Our pretraining comprises two stages with different datasets and resolutions, progressing from fixed low-resolution inputs to diverse resolutions and native aspect ratios. This setup allows the model to learn

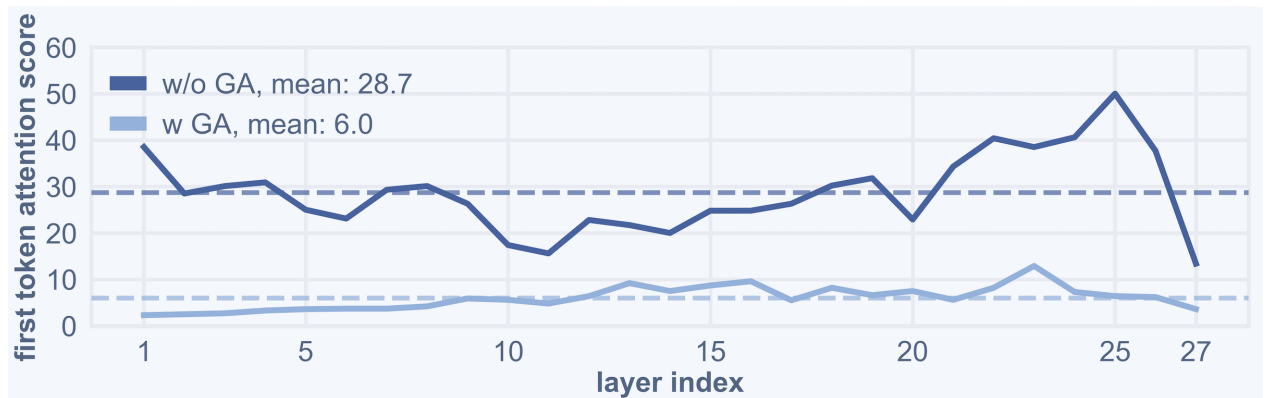


Figure 3 We simply plot the attention distribution of the first token in the input sequence, which is severe attention sink token. Without gated attention, the first token absorbs most of the attention mass.

foundational visual and linguistic representations while keeping the overall computational cost manageable.

Fixed Resolution Pretraining. We first pretrain GenLIP on Recap-DataComp-1B [40], a large-scale dataset of 1 billion unique image-text samples collected from the web. During this stage, we use the fixed 224×224 resolution images to reduce computational cost while learning strong foundational visual representations. We train GenLIP for totally 8 billion samples in this stage, corresponding to 8 epochs over the dataset.

Diverse Resolution Adaptation. We further fine-tune fixed-resolution pretrained GenLIP on public open-source caption datasets, the caption subset of Infinity-MM (stage1) [27] and BLIP3o-Long-Caption [10], totally 37 million image-text samples with long captions and higher resolution. Different from higher resolution adaptation in previous works [53, 68], we process images in their native aspect ratios, and resize them to keep the number of vision tokens within [16, 1024]. In this adaptation stage, we train GenLIP for only 1 epoch over the datasets. This stage helps the model adapt to variable-resolution inputs and learn finer-grained visual representations from dense text description, which is important for downstream tasks that require detailed visual understanding and precise image-text grounding.

Regularization. We apply two regularization techniques during GenLIP pretraining for effectively training deeper networks: layer scale and drop path. These two techniques are mainly used to stabilize training and prevent divergence when training deeper models, but found less impact on the final GenLIP performance.

Table 1 Overview of GenLIP configurations and pretraining setup. **Left:** model configurations. **Right:** two-stage pretraining details.

Model configurations.						Two-stage pretraining details.					
Model	Params	Layers	Dims	Heads	FFN-w	Stage	Dataset	Size	Resolution	Patches	Samples
GenLIP-L	0.3B	24	1024	16	2816	S1	Recap-DataComp-1B	1.0B	224	196	8.0B
GenLIP-So	0.4B	27	1152	16	3072	S2	Blip3o-Long-Cap	27M	AnyRes	16-1024	37M
GenLIP-g	1.1B	40	1536	24	4096		Infinity-MM-Stage1	10M			

Table 2 Hyperparameters and implementation details for GenLIP pretraining. “Batch Size” denotes the estimated global sample batch size.

Config	L/16	So/16	g/16
Optimizer	PyTorch AdamW		
Momentum	$\beta_1 = 0.9, \beta_2 = 0.95$		
Peak LR	1e-3		
Min LR	1e-6		
LR Decay	cosine decay		
Warmup Ratio	0.007	0.007	0.02
Gradient Clipping	1.0		
Max Packing Length	16384		
Batch Size	32K	32K	48K
Layer Scale	0.1		
Drop Path Ratio	0.1	0.1	0.2
Vocab Size	151936		
RoPE Theta	10000		

Pretraining Implementation. Table 2 summarizes the main pretraining hyperparameters of both two stages. We use the packing strategy to pack samples of variable lengths into long sequences with max length 16,384. The packed sequences are then batchified to improve training efficiency and hardware utilization. On top of this packing strategy, we implement exact per-sample Prefix-LM attention by the flex-attention in PyTorch, which allows variable sequence lengths and arbitrary attention masks. For image preprocessing, we use only resize and crop operations without additional augmentations on Recap-DataComp-1B [40].

There are three major differences between in the second stages: (i) the global batch size is reduced from 32K or 48K to 3.6K because the average sample length increases from 270 tokens to about 1200 tokens; (ii) the peak learning rate is reduced to $1e-4$; and (iii) transferred into native aspect ratio image processing. Besides, other training settings are kept the same as the first stage.

3.4 Discussion

Rather than introducing novel architectural components, GenLIP pursues the simplest possible paradigm for vision encoder pretraining, enabling seamless integration into MLLMs. Here, we summarize the key differences between GenLIP and prior works.

Differences from previous generative works. GenLIP differs from previous generative vision-language pretraining works [8, 23, 46, 67] in several key aspects: (i) Compared with VL-BEIT [8] and AIMv2 [23], GenLIP only learns from a single standard autoregressive language modeling objective, without masked image modeling or pixel reconstruction objective; (ii) Compared with CapPa [67], AIMv2 [23], and OpenVision2 [46], GenLIP discard additional text decoder and result into a minimalist modeling paradigm with a single unified transformer.

Differences from previous single Transformer pretraining works. GenLIP also differs from previous single Transformer pretraining works [19, 34]: (i) GenLIP focuses on pretraining a scalable vision encoder for modular MLLMs, rather than naive MLLMs; (ii) GenLIP is pretrained from scratch on caption datasets, while SAIL [34] and NEO [19] are trained by leveraging pretrained LLMs and large-scale instruction-tuning data; (iii) GenLIP improves attention implementation with a gated mechanism to make it better fit visual modeling as a vision encoder.

4 Experiments

To comprehensively evaluate the visual features learned by GenLIP, we begin with a “Let ViT Speak” test and then conduct extensive experiments on a broad suite of multimodal understanding benchmarks. We further analyze GenLIP’s scalability with respect to both data scale and model size. Finally, we provide ablations on key design choices, including the model architecture and the diverse-resolution adaptation stage.

4.1 Let ViT Speak

4.1.1 Direct Caption Generation

We begin with a simple but intuitive test of GenLIP’s generative ability by asking the model to describe an input image directly. We evaluate all three model scales on both common-image examples (Figure 4) and supplementary OCR-heavy examples reported in the appendix (Figure 8). For this test, we use temperature= $1e-6$, top_p=1.0, a maximum of 256 new tokens, and no beam search. Generation stops when the model outputs the end-of-sequence token. We use the simple prompt “Describe the image in details.” throughout.

As shown in Figure 4, GenLIP already produces fluent and semantically grounded descriptions. From stage 1 to stage 2, the responses become longer and more detailed, which is consistent with the finer-grained caption data used in the second pretraining stage. The captioning ability also improves with model scale. In the second example, the two smaller models, GenLIP-L16 and GenLIP-So16, mistake “Bulbasaur” for “Charmander”, whereas the largest model, GenLIP-g16, identifies it correctly and provides richer details.

4.1.2 Patch Semantics Readout

Beyond direct caption generation, we also probe what individual image patch embeddings represent by unembedding them into language tokens with the language modeling head. As shown in Figure 5, GenLIP spontaneously aligns some local visual regions with meaningful language concepts, an emergent property learned during pretraining. In the examples shown, both models associate selected regions with semantically relevant concepts ranging from natural objects to more abstract patterns. The GenLIP-g16-S2 model exhibits stronger alignment in both semantic correctness and relevance, likely due to the finer-grained captions and higher-quality images used in the second pretraining stage. Interestingly, this behavior is only observed in the



Figure 4 Let ViT Speak. We prompt GenLIP with “Describe the image.” and show representative generations. The first case compares three stage-1 models (GenLIP-L16-S1, GenLIP-So16-S1, and GenLIP-g16-S1) with one stage-2 model (GenLIP-L16-S2); the second case shows three stage-2 models. Green and red text indicate correct and incorrect key content, respectively.

two larger models, GenLIP-So16 and GenLIP-g16, with the latter showing more stable alignment. After stage 2, the readout semantics generally becomes more closely matched to the selected image regions. Although no explicit visual supervision is used, the model still learns to associate image patches with language concepts through generative pretraining on image-caption data.

Taken together, the caption generation and patch-semantics experiments show that GenLIP can jointly model and align visual and linguistic modality, supporting its use as a strong vision encoder for MLLMs.

4.2 Setup

4.2.1 Baselines

We compare our method, GenLIP, against a suite of representative vision-language pre-training models under multimodal understanding benchmarks. This includes contrastive methods such as CLIP [56], SigLIP [78], and SigLIP2 [68], as well as generative approaches like AIMv2 [23] and OpenVision2 [46]. For a fair comparison, all vision encoders are configured to produce the same number of visual tokens (patches). We use strong publicly available model variants for our baselines, such as ViT-L/14 for CLIP and AIMv2, and ViT-So/16 for SigLIP2. These methods are pretrained on substantially bigger training corpora (12.0B–40.0B image-text pairs) than GenLIP.

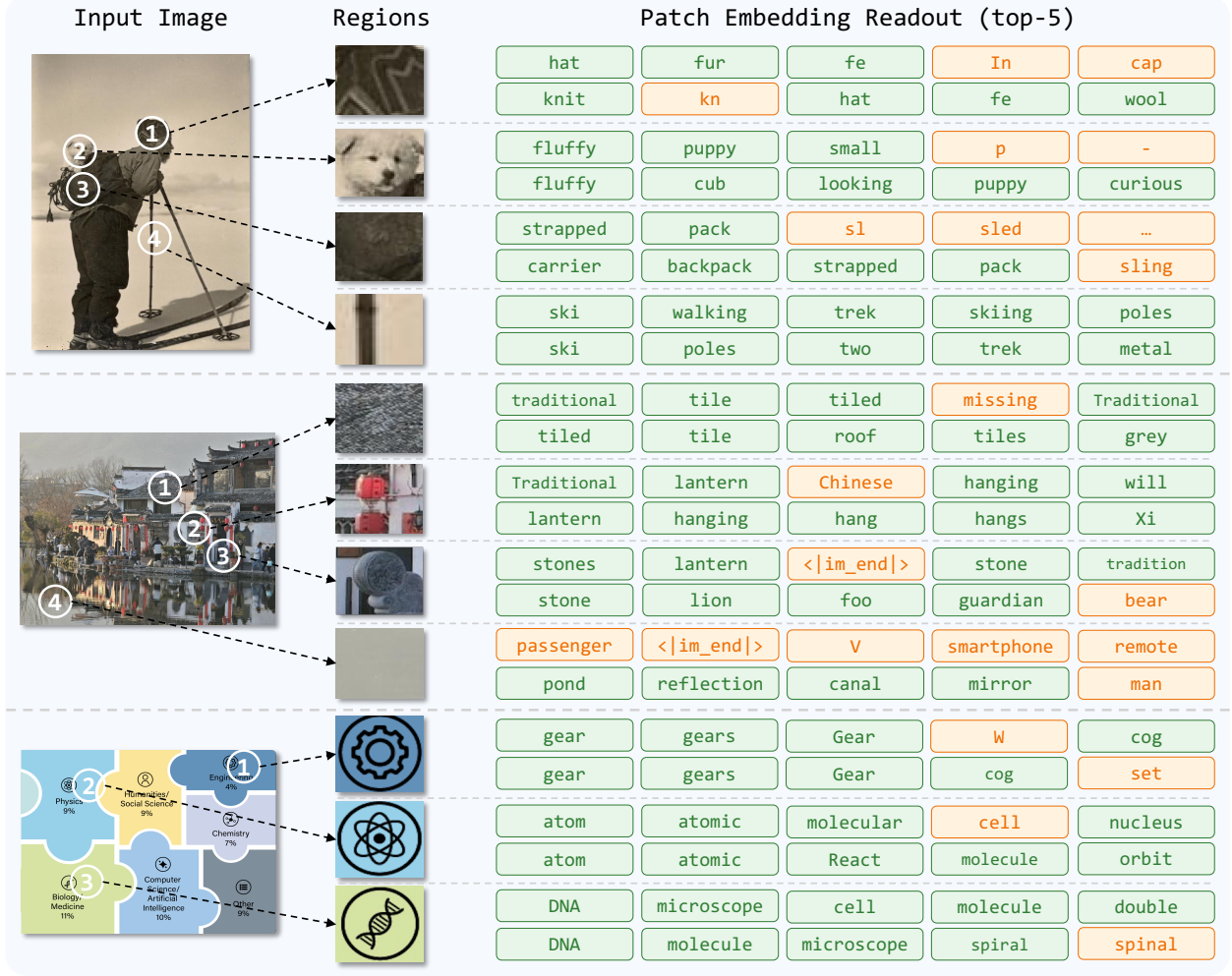


Figure 5 Patch Semantics Readout. We directly unembed selected image patch embeddings with the language modeling head to inspect the language concepts aligned with local regions. For each case, we show 3–4 regions for GenLIP-g16-S1 (top row) and GenLIP-g16-S2 (bottom row), together with the top-5 predicted tokens from left to right. Green boxes indicate related tokens and yellow boxes indicate unrelated ones.

4.2.2 Experimental Setup

Following Cambrian [65], we mainly adopt frozen visual representation evaluation, where the vision encoder is kept frozen and the language model is fine-tuned on downstream tasks. This protocol directly measures the quality of visual features learned by different VLP methods without the confounding effect of further fine-tuning the vision encoder. Based on the LLaVA-NeXT framework [44], we replace the original vision encoder with one pretrained by GenLIP or each baseline method, and then fine-tune the language model on an instruction tuning dataset. To better unleash the potential of the vision encoders, we replace the original 780K instruction-tuning set with the comprehensive LLaVA OneVision [35] dataset, which contains more than 3 million supervised fine-tuning (SFT) samples. We consider two LLM backbones of different sizes in our implementation, Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct [55], in place of the original LLM in LLaVA-NeXT.

4.2.3 Evaluation Benchmarks

To provide a comprehensive evaluation, we assess our method and all baselines across a diverse set of multimodal understanding benchmarks. These benchmarks are grouped into three categories to probe distinct capabilities:

document understanding and optical character recognition (Doc&OCR), general visual understanding (General VQA), and image captioning (Caption). All evaluations are conducted using the open-source LMMS-Eval toolkit [80].

Document and OCR. This category evaluates the model’s ability to recognize and interpret text within images, a critical skill for document analysis and scene text understanding. Following mainstream MLLMs [7, 35], we focus on a wide range of classic benchmarks, including ChartQA [50], OCRBench [47], InfoVQA [52], AI2D [32], TextVQA [59], DocVQA [51] and SEED-Bench-2-Plus [36].

General Visual Understanding. This group of tasks assesses the model’s broader capabilities in comprehending and reasoning about visual content. We employ four widely-used benchmarks, including MME [24], GQA [29], VQAv2 [26], and ScienceQA [48] for general VQA.

Image Captioning. To measure the model’s ability to generate descriptive text from images, we evaluate on NoCaps [2], COCO [49], and TextCaps [58] for evaluation. Performance is reported using the CIDEr metric.

For a holistic comparison, we report an overall average score across all 14 benchmarks (ALL AVG), computed as the mean of the per-benchmark scores. In particular, we rescale MME-P scores to the range [0, 100] based on the original score by 2000 (the maximum score for this subset), ensuring comparability.

4.3 Main Results

We provide all frozen visual representation evaluation results on multimodal understanding benchmarks in Table 3 and Table 4. Besides, we also provide results under the standard unfrozen LLaVA-NeXT evaluation setting in Table 5.

Table 3 Frozen visual representation evaluation under LLaVA-NeXT-Qwen2.5-1.5B. We test GenLIP models across three scales against baseline methods. The benchmarks are grouped into Doc&OCR, General VQA, and Caption tasks. “Arch” stands for “Model Architecture”, while “Data” denotes “Pretraining Data Scale”. “OpenVision2” is abbreviated as “OVision2”.

Model	Arch	Data	Doc&OCR							General VQA				Caption			ALL AVG
			ChartQA	OCR-B	DocVQA	TextVQA	AI2D	InfoVQA	SEED-2	VQAv2	GQA	SQA	MME-P	NoCaps	COCO	TextCaps	
CLIP [56]	L/14	12.8B	24.8	23.7	38.9	43.9	64.5	30.2	47.8	46.1	39.8	75.3	1218	55.5	72.5	117.9	53.1
AIMv2 [23]	L/14	12.0B	26.3	25.2	37.7	47.2	64.2	29.3	47.3	48.1	43.9	76.2	1157	80.1	73.6	122.4	55.7
OVision2 [46]	L/16	12.8B	30.7	45.6	43.3	49.2	65.6	28.1	47.8	44.0	42.7	75.5	1230	84.3	76.3	127.4	58.7
SigLIP [78]	L/16	40.0B	30.2	41.0	47.3	36.0	66.4	27.8	47.9	41.3	41.7	76.7	1203	84.0	76.1	120.7	56.9
SigLIP2 [68]	L/16	40.0B	33.4	45.7	45.1	50.3	66.7	28.2	45.7	43.1	42.6	76.9	1165	82.9	74.6	127.8	58.7
GenLIP	L/16	8.0B	41.2	51.1	51.1	53.6	66.6	30.7	51.1	44.4	41.5	76.1	1258	82.6	76.0	131.4	61.5
SigLIP2 [68]	So/16	40.0B	35.2	47.2	46.4	53.3	67.0	28.0	50.3	46.5	43.5	77.1	1220	84.3	77.1	131.5	60.6
GenLIP	So/16	8.0B	40.8	51.5	51.9	55.2	67.2	31.9	52.3	46.5	44.0	76.0	1215	87.5	81.5	129.5	62.6
SigLIP2 [68]	g/16	40.0B	35.3	47.3	47.6	54.7	66.7	29.7	49.6	50.1	45.2	76.2	1284	84.4	76.2	134.5	61.5
GenLIP	g/16	8.0B	45.0	55.6	57.0	59.0	68.9	33.9	53.3	49.1	45.5	77.5	1256	88.3	82.0	135.4	65.2

4.3.1 Frozen Feature Analysis

As presented in Table 3, GenLIP demonstrates strong performance across three model scales. Despite using fewer pretraining pairs, GenLIP achieves consistent gains over all baselines, including the 40B-pair pretrained SigLIP2. Under the Qwen2.5-1.5B setting, GenLIP improves the overall average (ALL AVG) over SigLIP2 by 2.5, 2.0, and 3.7 points at the L/16, So/16, and g/16 scales, respectively. The gains are especially pronounced on Doc&OCR benchmarks, which demand fine-grained document understanding and text-centric visual reasoning. Averaging over the seven Doc&OCR tasks in Table 3, GenLIP achieves 49.3, 50.1, and 53.2 at L/16, So/16, and g/16, outperforming SigLIP2 by 4.3, 3.3, and 5.9 points, respectively.

This advantage remains under a larger LLM. As shown in Table 4, scaling the LLM to Qwen2.5-7B yields consistent trends with the Qwen2.5-1.5B setting. Under this setting, GenLIP outperforms SigLIP2 by 2.4 and

Table 4 Frozen visual representation evaluation under LLaVA-NeXT-Qwen2.5-7B. Except for the LLM size, all settings are the same as those used in LLaVA-NeXT-Qwen2.5-1.5B.

Model	Arch	Data	Doc&OCR							General VQA				Caption			ALL AVG
			ChartQA	OCR-B	DocVQA	TextVQA	AB2D	InfoVQA	SEED-2	VQAv2	GQA	SQA	MME-P	NoCaps	COCO	TextCaps	
CLIP [56]	L/14	12.8B	36.6	29.6	48.4	52.6	76.3	39.0	55.1	49.4	39.6	85.2	1316	63.1	54.4	127.9	58.8
AIMv2 [23]	L/14	12.0B	36.8	30.9	46.6	54.5	76.9	37.5	55.1	44.0	37.9	85.2	1240	66.5	55.9	130.5	58.6
OVision2 [46]	L/16	12.8B	42.5	49.9	49.5	58.8	78.4	33.8	53.8	60.0	47.2	85.9	1325	79.4	69.6	133.8	64.9
SigLIP [78]	L/16	40.0B	41.7	45.7	50.5	56.0	79.3	34.8	55.8	57.8	46.2	86.7	1275	81.5	72.0	131.1	64.5
GenLIP	L/16	8.0B	52.7	59.2	61.7	62.9	80.4	38.8	59.0	56.4	51.3	85.4	1320	81.1	71.3	139.4	69.0
SigLIP2 [68]	So/16	40.0B	46.6	55.6	56.3	63.5	81.3	37.2	56.4	64.5	52.2	87.1	1422	84.1	76.4	139.3	69.4
GenLIP	So/16	8.0B	55.3	63.5	66.3	65.7	81.0	41.4	60.8	60.5	52.4	86.4	1424	83.1	74.8	142.1	71.8
SigLIP2 [68]	g/16	40.0B	47.2	55.6	56.3	63.5	81.0	36.4	56.4	62.7	49.3	87.7	1422	82.0	72.3	142.7	68.9
GenLIP	g/16	8.0B	57.1	65.9	69.0	66.8	81.0	43.6	61.1	64.4	54.5	87.0	1483	85.0	75.5	144.8	73.6

4.7 points on average score at the So/16 and g/16 scales, respectively. Similar to the Qwen2.5-1.5B setting, GenLIP consistently performs best on Doc&OCR benchmarks, highlighting its strong fine-grained visual-text alignment.

Across both frozen settings, GenLIP not only surpasses contrastive VLMs such as CLIP [56] and SigLIP [78], but also outperforms prior encoder-decoder generative VLMs, including AIMv2 [23] and OpenVision2 [46]. These generative baselines use an additional text decoder for language modeling, and OpenVision2 is further pretrained on the stronger Recap-DataComp-1B v2 corpus with a longer training schedule. Overall, the results suggest that GenLIP’s minimalist architecture and objective can yield stronger visual representations with improved data efficiency.

We also observe that GenLIP scales favorably with model size, while SigLIP2 shows comparatively smaller gains when scaling up. These results support two hypotheses: (i) simplifying both the architecture and the objective can enable more efficient scaling; and (ii) larger model capacity helps GenLIP learn both broad visual knowledge and fine-grained alignment for multimodal understanding.

4.3.2 Standard LLaVA-NeXT Evaluation

We further evaluate GenLIP under the standard LLaVA-NeXT setting following prior work [73], where the vision encoder is unfrozen and fine-tuned jointly with the language model during instruction tuning. As shown in Table 5, GenLIP performs strongly under two fixed patch budgets and achieves competitive overall results across both Doc&OCR and General VQA tasks. GenLIP shows consistent advantages on Doc&OCR benchmarks.

Taken together, both the frozen and standard evaluations indicate that GenLIP provides strong and consistent performance across diverse multimodal understanding tasks, including Doc&OCR, General VQA, and captioning. In particular, GenLIP consistently excels on Doc&OCR tasks, which demand fine-grained visual recognition and precise visual-text alignment.

Overall, these results indicate that GenLIP, a simple yet effective generative vision-language pretraining method, can learn rich and versatile visual representations for multimodal understanding with high data efficiency. Compared with more complex alternatives (e.g., SigLIP2 with larger pretraining corpora and more elaborate training recipes), GenLIP exhibits highly competitive and often achieves better downstream performance. This suggests that minimalist generative vision-language pretraining is a promising direction for learning strong, scalable visual representations for MLLMs.

4.4 Scalability Analysis

To investigate the detailed scaling pattern of GenLIP, we discuss both data and model scalability of GenLIP, which are two key factors for VLP pretraining.

Table 5 Multimodal understanding results under standard LLaVA-NeXT settings. All models are evaluated using identical configurations: the same data and LLM and anyres image processing configuration [44].

Patches	Model	Arch	Data	Doc&OCR						General VQA						ALL AVG
				ChartQA	DocVQA	TextVQA	OCR-B	LiveVQA	AI2D	MMBench	MME-C	MME-P	POPE	RWQA	MMStar	
576	CLIP [56]	L-14	12.8B	75.2	66.5	62.5	52.5	47.4	73.2	74.6	48.0	75.6	88.8	63.7	49.0	64.8
	MLCD [4]	L/14	12.0B	76.5	67.8	61.7	53.1	48.4	77.0	76.5	54.1	79.9	88.7	61.1	51.0	66.3
	AIMv2 [23]	L/14	12.8B	77.2	72.7	65.9	57.2	47.3	75.4	78.6	48.3	75.0	88.4	62.2	50.2	66.5
	RICE-ViT [73]	L/14	13.0B	79.2	72.3	65.9	57.5	48.9	77.9	76.6	54.6	80.7	88.5	63.1	51.8	68.1
	GenLIP	So/16	8.0B	79.3	75.2	68.5	59.7	48.4	78.6	77.7	48.6	78.2	89.2	65.9	53.1	68.5
729	SigLIP [78]	So/14	40.0B	76.7	69.3	64.7	55.4	48.4	76.2	77.0	46.1	79.9	88.8	63.7	47.3	66.1
	SigLIPv2 [68]	So/14	40.0B	79.1	70.2	66.2	58.7	48.6	77.0	77.1	46.6	80.4	89.3	63.4	52.8	67.5
	RICE-ViT [73]	L/14	13.0B	82.6	75.1	66.2	58.8	49.5	76.5	77.6	54.1	79.0	89.1	62.9	51.2	68.6
	GenLIP	So/16	8.0B	83.0	76.9	69.6	64.7	50.4	79.1	78.1	54.5	80.1	89.4	65.1	53.2	70.3

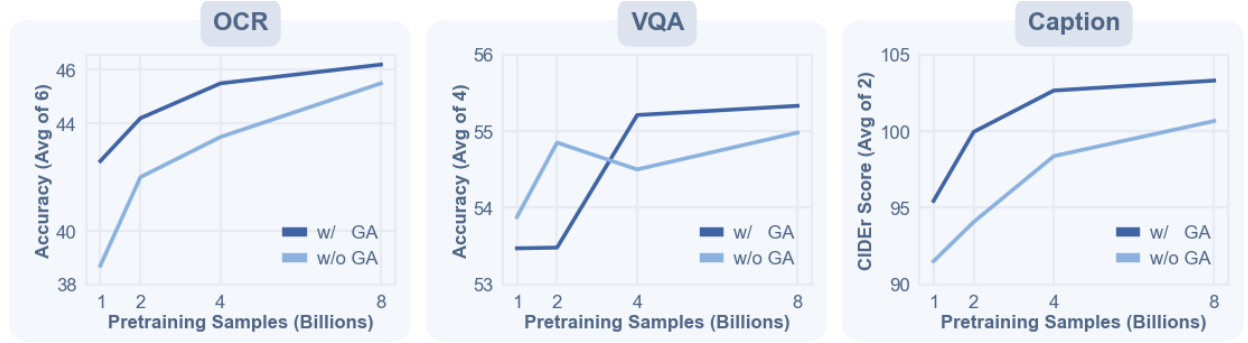


Figure 6 Data Scaling Behavior. Performance on three kinds of tasks as the number of pretraining samples in the first stage is scaled from 1.0B to 8.0B. We report and plot the curve of the average score for Doc&OCR, VQA, and Caption tasks. The x-axis in each subplot corresponds to the pretraining data scale.

4.4.1 Data Scaling

We first study data scaling in Fig. 6, where we pretrain GenLIP (with or without gated attention) on Recap-DataComp-1B with different numbers of training samples, ranging from 1.0B to 8.0B. As the data scale increases from 1.0B to 8.0B, GenLIP shows sustained improvements on multimodal understanding benchmarks. We observe steeper gains when scaling from 1.0B to 4.0B, while the improvement curve becomes flatter when further scaling to 8.0B. In particular, the average performance on VQA and caption tasks shows only minor improvements when scaling from 4.0B to 8.0B. Based on this trend, we use 8.0B samples as the default data scale for GenLIP pretraining in our main results.

4.4.2 Model Scaling

We also investigate how GenLIP performance changes with model size by pretraining GenLIP at the L/16, So/16, and g/16 scales. Besides the final results after diverse resolution adaptation shown in Table 3, we additionally provide results for models pretrained only with fixed low resolution on Recap-DataComp-1B in Table 6. Across both pretraining stages, GenLIP shows consistent performance gains with increasing model size. An important observation is that GenLIP-L/16 lags behind GenLIP-So/16 and GenLIP-g/16 only with fixed low-resolution pretraining, while the gap between g/16 and So/16 is relatively small. This suggests that an appropriate model size is important for GenLIP to learn strong visual representations and better performance on downstream tasks.

4.5 Ablations

Table 6 Frozen visual representation evaluation of GenLIP pretrained at different model scales across two stages. “S1” and “S2” denotes the pretraining stage 1 and 2 respectively.

Model	Arch	Doc&OCR							General VQA				Caption			ALL AVG
		ChartQA	OCR-B	DocVQA	TextVQA	AI2D	InfoVQA	SEED-2	VQAv2	GQA	SQA	MME-P	NoCaps	COCO	TextCaps	
SigLIP2	L/16	33.4	45.7	45.1	50.3	66.7	28.2	45.7	43.1	42.6	76.9	1165	82.9	74.6	127.8	58.7
GenLIP-S1	L/16	34.3	28.9	44.5	43.0	64.5	28.5	49.1	44.0	41.9	75.2	1136	77.3	71.0	114.1	55.2
GenLIP-S2	L/16	41.2	51.1	51.1	53.6	66.6	30.7	51.1	44.4	41.5	76.1	1258	82.6	76.0	131.4	61.5
SigLIP2	So/16	35.2	47.2	46.4	53.3	67.0	28.0	50.3	46.5	43.5	77.1	1220	84.3	77.1	131.5	60.6
GenLIP-S1	So/16	37.6	39.2	49.8	50.5	65.3	29.7	51.3	45.4	43.8	75.2	1157	80.9	73.6	125.7	58.9
GenLIP-S2	So/16	40.8	51.5	51.9	55.2	67.2	31.9	52.3	46.5	44.0	76.0	1215	87.5	81.5	129.5	62.6
SigLIP2	g/16	35.3	47.3	47.6	54.7	66.7	29.7	49.6	50.1	45.2	76.2	1284	84.4	76.2	134.5	61.5
GenLIP-S1	g/16	34.6	42.5	53.7	53.1	65.5	29.6	51.1	45.3	43.5	75.9	1164	82.0	74.0	132.0	60.0
GenLIP-S2	g/16	45.0	55.6	57.0	59.0	68.9	33.9	53.3	49.1	45.5	77.5	1256	88.3	82.0	135.4	65.2

4.5.1 Comparison with Other VLPs

A key property of GenLIP is data efficiency: as shown above, GenLIP pretrained on 8B pairs can surpass baselines pretrained with substantially larger corpora. To further validate this property, we conduct a controlled comparison among a contrastive method (SigLIP), an encoder-decoder generative method (OpenVision2), and our GenLIP under the same pretraining data budget.

Specifically, we train SigLIP, OpenVision2, and GenLIP on the same 2.0B samples from Recap-DataComp-1B. For GenLIP, we run only the first pretraining stage and evaluate directly at a 384×384 input resolution. For SigLIP and OpenVision2, we pretrain at 224×224 and further conduct a short high-resolution adaptation stage at 384×384 for 0.2B samples. For SigLIP, we implement the vanilla sigmoid contrastive loss without additional tricks from SigLIP2 [68].

We evaluate frozen visual representations of these methods under the same protocol in Table 7. Under the same data budget, GenLIP still achieves strong performance on both Doc&OCR and General VQA tasks. While GenLIP outperforms the baselines on most benchmarks, it trails OpenVision2 on OCRBench by 6.3, which is likely related to the absence of high-resolution adaptation in GenLIP under this controlled setting and the known difficulty of dense-text recognition with low-resolution pretraining.

Overall, this controlled comparison supports that our minimalist generative VLP method can be more data-efficient than both contrastive and prior generative alternatives.

Table 7 Ablation between different pretraining methods.

Model	Arch	Data	OCR							General VQA					
			ChartQA	OCR-B	DocVQA	TextVQA	AI2D	InfoVQA	SEED-2	AVG	VQAv2	GQA	SQA	MME-P	AVG
SigLIP	So/16	2.0B	26.1	36.2	38.6	44.3	64.2	25.8	46.0	40.2	42.7	39.8	75.1	1132	53.6
OVison2	So/16	2.0B	27.8	43.2	41.2	44.7	64.1	26.8	46.3	42.0	44.2	40.3	74.8	1158	54.3
GenLIP	So/16	2.0B	35.0	36.9	46.0	47.1	64.9	29.3	50.3	44.2	45.4	42.0	75.6	1156	55.2

4.5.2 Gated Attention

In Fig. 6, we plot data scaling curves of GenLIP with and without gated attention, showing consistent advantages of gated attention across data scales. Gated attention improves data efficiency, especially in the low-data regime, where the variant with gated attention achieves higher performance than the one without. It also leads to better convergence and improves the final performance by a notable margin.

4.5.3 Native-Aspect-Ratio Adaptation

We evaluate GenLIP pretrained with two stages under different evaluation resolutions, which validates the effectiveness of the native-aspect-ratio adaptation stage. To test the model’s behavior under different input resolutions, we evaluate frozen visual representations of GenLIP (after each stage) across multiple resolutions under the same protocol as in Table 3 (Fig. 7).

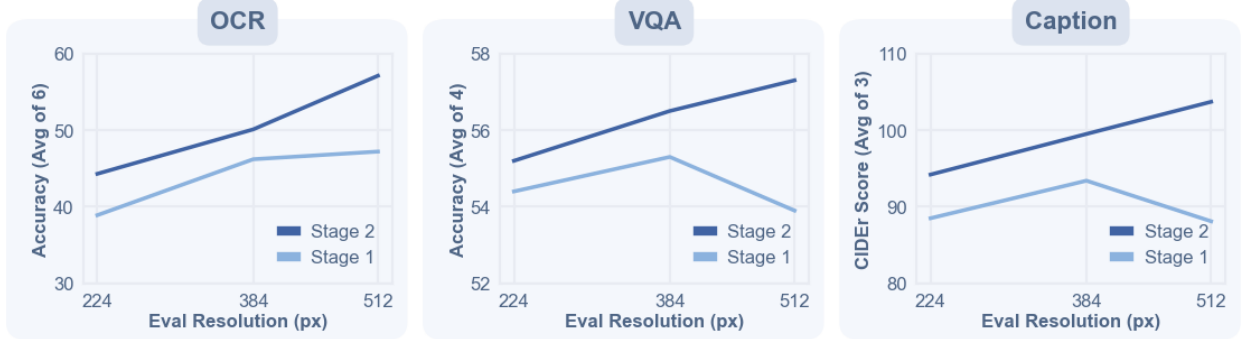


Figure 7 Validation of Native Aspect Adaptation. We evaluate the frozen visual representation of GenLIP-So/16 pretrained after two stages on the same setting as shown in Table 3. The x-axis corresponds to the input resolution in evaluation, and the y-axis corresponds to the average score on OCR, VQA and Caption tasks, respectively.

4.6 Discriminative Ability

To assess the discriminative quality of GenLIP’s visual representations, we adopt the frozen-backbone evaluation protocol from DINOv2 [53] and probe the frozen visual features on ImageNet-1K [17] for classification and ADE20K [82] for semantic segmentation. Because GenLIP has no [CLS] token, we use attentive probing for classification, and we use only a linear layer on patch features for dense probing on ADE20K.

Table 8 Frozen feature evaluation on the ImageNet-1K and ADE20K validation set. We report top-1 accuracy on ImageNet-1K and mIoU on ADE20K. No test-time augmentation used in evaluation. “w/o GA” denotes the variant without introducing gated attention.

Method	Arch	ImageNet-1K	ADE20K
CLIP	L/14	85.1	39.0
SigLIP	So/14	86.7	40.8
SigLIP2	So/14	88.9	45.4
GenLIP w/o GA	So/16	76.2	-
GenLIP	L/16	83.9	41.0
GenLIP	So/16	84.3	42.8
GenLIP	g/16	85.2	44.5

As shown in Table 8, GenLIP learns decent transferable discriminative visual features without explicit visual supervision. There are two related findings: (i) gated attention effectively alleviates the degraded discriminative representations due to attention sink, (ii) the discriminative ability scales with GenLIP model sizes. The biggest variant of GenLIP, GenLIP-g/16, achieves 85.2 top-1 accuracy on ImageNet-1K and 44.5 mIoU on ADE20K with frozen representations. Notably, GenLIP outperforms the pure contrastive methods CLIP and SigLIP on ADE20K under the same model sizes, but lags behind SigLIP2 which introduces dense supervision [68]. Overall, this result demonstrates our pretraining method delivers competitive visual representations for discriminative tasks with an extremely simple pretraining method.

Additional qualitative examples, evaluation details, and a detailed discussion of attention sink are provided in the appendix.

5 Conclusions

This work presents GenLIP, a minimalist generative vision-language pretraining method by a simple unified transformer architecture and a simple language modeling objective. Begin with a single transformer that jointly models both visual and textual inputs, GenLIP aligns the visual and textual modality in an early fusion way with a single generative objective. Despite its architectural and objective simplicity, GenLIP demonstrates remarkable data efficiency and scalability for vision-language pretraining, with relatively less training data to achieve competitive or superior performance across a wide range of multimodal benchmarks. We hope our exploration of generative vision-language pretraining will inspire future research toward more effective and scalable multimodal learning.

Limitations. Several limitations warrant consideration: (i) our validation experiments are conducted on an academic-scale MLLM setting, LLaVA-NeXT, and the generalizability to cutting-edge ones remains to be verified; (ii) the pretraining dataset is limited to 1.0B scale, the scaling behavior at even larger volumes is yet to be explored; (iii) the reliance on high-quality captions introduces significant data acquisition costs.

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Appendix

A Supplementary Qualitative Results

We provide additional qualitative results that complement the “Let ViT Speak” analysis in Sec. 4.1. Besides illustrating the strengths of GenLIP, these cases also expose its remaining failure modes on challenging detail-sensitive inputs.

In Figure 8, we further evaluate GenLIP on challenging OCR-heavy examples. These three cases test (a) receipt understanding, (b) geometric-shape counting and placement, and (c) recognition of tiny characters and numbers. All three model variants show non-trivial OCR ability, although clear errors remain:

(a) In the first case (Figure 8(a)), GenLIP-L16-S2 recognizes most characters but fails on the long number sequences (Tax Id and IBAN) and the two tables. GenLIP-So16-S2 encounters similar difficulties and produces repeated output. In contrast, GenLIP-g16-S2 reads out the table structure much more accurately, missing only one number and the word “Opener”.

(b) In the second case (Figure 8(b)), GenLIP-L16-S2 and GenLIP-So16-S2 make mistakes in both the number and placement of geometric shapes. GenLIP-g16-S2 is substantially more accurate, with the main remaining error being that it identifies the acute triangle in the bottom row as a right triangle.

(c) In the last case (Figure 8(c)), GenLIP-L16-S2 fails to detect the number on the plane, and GenLIP-So16-S2 outputs the wrong number. GenLIP-g16-S2 identifies the number correctly but still makes a spatial error.

Overall, these examples show that GenLIP already acquires meaningful OCR ability even without an OCR-specific pretraining corpus. This ability scales clearly with model size: larger models recognize and describe subtle details more accurately. At the same time, the observed errors show that long number strings, precise spatial layouts, and tiny text remain challenging. These cases help explain both the strong Doc&OCR performance of GenLIP and the residual gaps that remain in detail-sensitive settings.

In Figure 9, we provide four more cases in addition to Figure 5 using the same model configurations.

B Additional Implementation Details

Frozen Visual Representation Evaluation. We summarize the training settings for frozen visual representation evaluation. Relative to the default LLaVA-NeXT [44] setup, we make three modifications: (i) we replace the original LLM LLaMA3-8B with Qwen2.5 models; (ii) we replace the original 780K SFT dataset with the 3M SFT dataset from LLaVA-OneVision; and (iii) we use the simplest image preprocessing pipeline, consisting only of resize and crop operations, without “anyres” processing designed for high-resolution images. All other training settings remain unchanged, including the optimization hyperparameters, batch size, iterations, and the 2-layer MLP projector.

Metric Aggregation. For the frozen visual representation results in Sec. 4.2, we report ALL AVG as the unweighted mean over all 15 benchmarks. Because MME-P is reported on a 0–2000 scale, we divide it by 2000 and map it into the range [0, 100] before averaging, so that it is numerically comparable with the other metrics. And the CIDEr scores on caption benchmarks are already in a normal range, so we keep them unchanged in ALL AVG calculation.

Pretraining Implementation. The main hyperparameters of GenLIP pretraining are summarized in Table 2. Stage 1 uses fixed 224×224 inputs and trains for 8B samples to learn strong foundational visual representations. Stage 2 then adapts the model on higher-resolution caption data with native aspect ratios, resizing each image so that the number of visual tokens stays within [16, 1024]. For efficiency, we pack variable-length samples into sequences with a maximum length of 16,384 tokens and implement exact per-sample Prefix-LM masking with PyTorch flex-attention. Because the second stage contains much longer sequences on average, its global batch size is reduced accordingly, while the remaining optimization settings follow Stage 1.

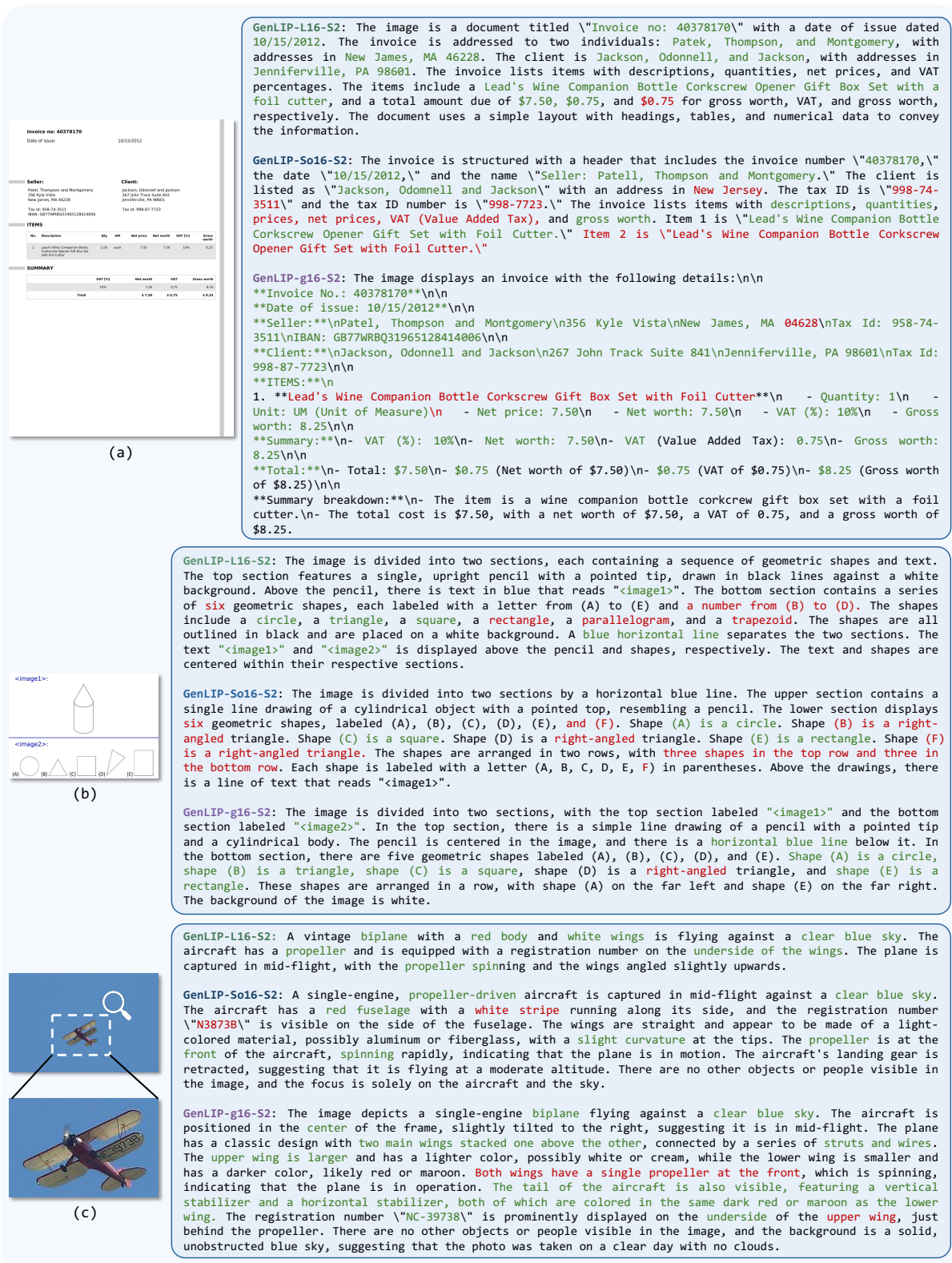


Figure 8 Additional OCR Cases. Representative GenLIP generations on three challenging examples that require fine-grained detail recognition.

Input Image	Regions	Patch Embedding Readout (top-5)				
		tree	G	F	S	bons
		tree	pine	needle	bons	Pak
		Carving	carved	H	X	graffiti
		characters	chinese	Hang	East	Mong
		< im_end >	a	making	high	matrix
		mountains	steep	peaks	rocky	mountain
		safety	le	suspension	stairs	st
		scaffold	scaff	bamboo	metal	caff
		houses	Houses	tiled	are	sh
		steep	rooft	pitched	roofs	roof
		black	skate	grey	number	bicycle
		helmet	helmets	Helm	protective	motorcycle
		cable	t	tram	contrasting	us
		cable	t	tram	street	conductor
		entirely	train	gray	< im_end >	Train
		"?"	(?)	"?"	question	questioning
		Times	TIMES	The	Press	New
		Times	Press	MS	Best	TIMES
		BEST	Best	1	I	New
		1	Best	BEST	#	I

Figure 9 Additional Patch Semantics Cases. Further examples of direct semantic readout from image patch embeddings for GenLIP-g16-S1 and GenLIP-g16-S2. The stage-2 model generally shows stronger alignment.

C Discussion: Attention Sink and Gated Attention

In GenLIP, we observe the “attention sink” phenomenon, which has also been reported in prior transformer studies in both vision [16] and language [54, 72]. At a high level, attention sink arises from the sum-to-one normalization of softmax attention: for each query token, the model must distribute a fixed unit mass over all keys. In practice, this often encourages the network to allocate a disproportionate amount of attention to a small subset of tokens that behave like persistent “registers” and absorb information from many other positions.

The manifestation of attention sink depends on the attention pattern of the modality. In vision transformers with bidirectional self-attention, sink behavior often appears as a small number of tokens in low-semantic regions that attract attention from many other visual tokens [16] and exhibit unusually high norm. In contrast, in autoregressive language models the phenomenon is typically more structured: early tokens, especially the first token, tend to receive disproportionately large attention weights from subsequent positions regardless of content. As discussed in StreamingLLM [72], such sink tokens may preserve useful global context information and can even be exploited for efficient long-context inference. This difference is largely explained by the underlying attention mechanism: full attention in vision does not privilege a fixed position a priori, whereas causal attention in language naturally makes early tokens accessible to all later tokens and therefore encourages early ones to serve as shared context carriers.

The Prefix-LM attention used in GenLIP combines bidirectional attention over the visual prefix with causal attention over the text suffix, making its sink behavior closer to that of autoregressive language models. The input sequence follows the organization $[v_0, \dots, v_M, t_0, \dots, t_L]$, positioning visual tokens as the prefix for text generation. Because the loss is backpropagated only through text tokens, the model tends to compress information useful for generation into a few preceding visual tokens that are broadly accessible to the text tokens. Under this structure, the first visual token v_0 becomes a particularly favorable sink candidate, since it can be attended by all subsequent text tokens and thus can act as a compact carrier of global visual context.

Empirically, we find that this behavior can partially degrade the discriminative quality of the visual representation, as reflected by the degraded linear-probing results of the “w/o GA” variant in Table 8. This observation motivates the introduction of gated attention in GenLIP, which alleviates overly concentrated sink behavior and improves the quality of the learned visual features. We also note that many encoder-decoder generative VLP architectures are less affected by this issue. Because the visual encoder and text decoder are separated, sink behavior is largely confined to the decoder side and therefore has much weaker direct impact on the quality of the visual encoder representations.